

Lec 46 Denoising Diffusion Probabilistic Model-Forward Process

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Transcript

[00:00] Hello everyone. Welcome back to the lecture series on fundamentals of generative AI and large language models. In my previous session, we have discussed about the mathematical foundations that are required for understanding diffusion probabilistic models. In today's session, we will discuss about denoising diffusion probabilistic model and we will look into only the forward diffusion process, okay? So, we'll start with what is denoising diffusion probabilistic model, okay?

[00:34] So, denoising diffusion probabilistic model, or we call it as DDPM, it's a specific formulation of diffusion probabilistic model, okay? So, it was um uh it it came into picture through the paper denoising diffusion probabilistic models by Jonathan Ho and uh the team from UC Berkeley in 2020. So, we will see what is this DDPM and how in this DDPM, how the forward process happens, okay? So, we all know that in any diffusion probabilistic model, we have the forward diffusion process and the reverse diffusion process, okay? So, in the forward diffusion process, we start from

a clean image and then we keep on adding noise subsequently for a number of steps till we get almost pure noise, okay? So, this is

[01:34] cleaner image clean image to complete noise conversion that we do in the forward diffusion process. And in the reverse reverse diffusion process, we start from almost pure noise and then go through several steps till we come to the clean image, okay? So, this is known and this is fundamental of any diffusion probabilistic model. Now, we will see what is the mathematical formulation for forward diffusion model in denoising diffusion probabilistic models, okay?

[02:08] Now, in one of my previous lectures, I have introduced you all to what is Markov chain. Markov chain is basically a Markov chain states that what will be my next state depends only on my current state, not on all the steps that have led me to the current step, okay? So, that is Markov chain, and our forward diffusion process is, you know, modeled as a Markov chain. So, we have real data, which is our image, then it is run through T steps of denoising, and every step is more noisy than the previous step, okay? So, this is this we have seen. Every noise Every step, when T equal to 1, 2, that time the noise added to the image is much lesser compared to when we have T equal to 30 or 35 or, you know, 40. So, as and

[03:12] when the time progresses, the time steps progresses, the amount of noise that is added to the image keeps on increasing, and finally we get a complete noise noisy image, okay? So, if we have x_{t-1} this image, x_t will become this one. And this is the conversion. There is noise addition from x_{t-1} to x_t , okay? » [snorts] » $Q(x_t | x_{t-1})$ given x_{t-1} is the probability of getting x_t given x_{t-1} . So, what is the probability that I get x_t given I know that this is my x_{t-1} . This is given to me.

[03:56] So, what is the probability of getting a noisier sample from a sample which is less noisier, okay? And that is given by this probability distribution, okay? Now, this x_t is sampled from a Gaussian distribution with μ mean μ and covariance σ . Now, when I tell this one, because the forward process, it's a fixed process, it is there is nothing uh no learning happens in the forward process. We consider it as a predefined Gaussian process Gaussian distribution.

[04:34] The probability distribution is uh can be thought of a Gaussian distribution with μ and covariance σ , you know, σ . And it is written as $p(x_t | x_{t-1})$ within bracket, this is x_t , which is your sample that you have sampled and this is μ is your mean and σ is the covariance. So, that is how mathematically we write uh uh or we represent when a sample x_t is taken or x_t is sampled from a Gaussian distribution with mean μ and covariance σ . So, that is the mathematical way of writing.

[05:12] Now, we will look into more details into the maths that is involved in the forward diffusion process. So, we know this is the probability uh distribution of x_t given x_{t-1} equal to this $p(x_t | x_{t-1})$. x_t is sampled from this distribution and this is the mean and this is the covariance. Okay? Now, this is forward diffusion distribution,

as I told. This is this term is the covariance term. In forward diffusion process, we have this equation, okay? Now, if you see what exactly this covariance is giving, the covariance describes how uncertainty is distributed across the dimensions and how different dimensions are related, okay? So, we have in our previous lecture, we have discussed what exactly covariance is and here we are using that

[06:15] covariance, you know, to understand the uncertainty associated with the noise across the dimensions and how different dimensions are correlated, okay? So, this is just once more I have written the covariance term can be written if there are two variables X_1 and X_2 , this is written as, you know, variance variance of X_1 , covariance between $X_1 X_2$, covariance between $X_2 X_1$, variance of X_2 . This is for two variables. X_1 and X_2 , so $X_1 X_2$. If it is like that, if your X is like this, okay? And zero covariance means there is no relationship between the dimensions, okay? This we will keep it in mind.

[07:05] Now, in this covariance term, we see it is written $\beta_t I$, okay? Where β_t is the magnitude of the noise, okay? We have seen in our previous discussion also, β_t was the was deciding how much noise will be added to to the forward diffusion process in a particular time step t , okay? And what is I ? I is basically the identity matrix, okay? It is written as like this. Now, $\beta_t I$, if we do a, you know, uh uh, you know, like this, this is a, you know, matrix and then we get $\beta_t I$. So, this is my $\beta_t I$, which is basically my covariance term. Okay?

[07:54] Now, in the forward diffusion process, again, I have written that same equation. This is the probability distribution of getting x_t given x_{t-1} or conditioned on x_{t-1} . So, your x_t is conditioned on what your x_{t-1} is having. How much your x_{t-1} one its step image, how much noise it is having. Based on that one, your x_t will be, you know, made. How some extra noise will be added to it. Now, this x_t is sampled from that distribution. This term is the mean because we know that is the way we write, you know, in terms of Gaussian distribution μx_t . This term becomes the mean and this is the covariance term.

[08:42] Now, if you see, once again, I have written x_t is the image at time step t . Uh, x_t is the image at time step t . t is missing, okay? Um, x_{t-1} is the image at time step $t-1$. I is the identity matrix and β_t controls how much noise is added. Okay? So, till this one, I hope it is clear. Now, given the previous image x_{t-1} , the next image x_t is sampled from Gaussian distribution with mean square root of $1 - \beta_t$ into x_{t-1} and covariance $\beta_t I$, $\beta_t I$.

[09:28] What exactly it means? That I have an image at the time step x at the time step t equal to at the time step $t-1$. Okay? So, how will I get x_t from this x_{t-1} ? So, once I have the previous image x_{t-1} , the next sample x_t , it is basically sampled from the Gaussian distribution with this this mean. This is the mean and this is the covariance. So, I have a Gaussian distribution having this mean

and this covariance β_t into I and mean square root of $1 - \beta_t$ x_t x_t minus 1 and there from there I sample x_t . Okay?

[10:15] Now, if you see, the mean is a scaled version of the previous image. What is What do I mean by that? This is x_t minus 1 basically, it is my previous image. So, this is my previous image. Correct? This is my previous image. So, x_t minus 1 into square root of $1 - \beta_t$, that is the mean. So, the mean is a scaled version of the previous image and the scaling factor is root over of $1 - \beta_t$. Where β_t , you know, varies between 0 and 1.

[10:55] Now, the next question is, why such a scaling of the signal strength is required? Is it required? Or if it is not there, what will happen? Okay? So, that we will see little later. Again, I have written the same equation here because we are This is the primary equation for the forward diffusion process, so I have kept it here. And then what I have done is, let α_t equal to $1 - \beta_t$. So, there is a substitution that I want to do here and I have rewritten this equation q_{x_t} given x_t minus 1 is equal to $N(x_t | \mu_t, \sigma_t^2)$ sample from this distribution with $1 - \beta_t$ is α_t . So, square root of α_t $x_t - 1$ and $1 - \alpha_t$ because this is β_t . So, uh α_t equal to $1 - \beta_t$. So, β_t becomes equal to

[11:52] $1 - \alpha_t$. So, $1 - \alpha_t$ into » [clears throat] » I. Okay. So, this is in terms of α_t . Now, a sample from here can be written the sample is x_t can be written as x_t equal to square root of α_t into $x_t - 1$ plus square root of $1 - \alpha_t$ into ϵ_t where ϵ_t belongs to a distribution with zero mean and variance one. Now, how I have written this one? Any sample can be written as you know, uh This This is basically your mean.

[12:35] And this is basically your covariance. Okay? So, it can be written as the mean plus standard deviation. From that uh idea, this has been written as the mean plus this is the standard deviation. Okay? And with ϵ_t uh which is uh which is which belongs to a distribution with Gaussian distribution with mean zero and variance one. Okay? Now, this is basically your scaled signal strength. That is how much of information was there in the image, a scaled version of that one is this this term and this is the noise term.

[13:21] Okay? Because ϵ_t is taken from a you know, random vector from Gaussian distribution, standard Gaussian distribution. Now, this is the signal is scaled with square root of α_t , okay? Now, what happens if the signal is not scaled? We will see that one, okay? So, here we have seen it is scaled to with square root of α_t . Now, if I have x_t equal to x_t minus one, which is my complete previous image, plus square root of one minus α_t into ϵ_t , then what happens? If I take variance of x_t , I can write equal to variance of x_t minus one plus variance of this term.

[14:07] So, I will write it here. Variance of x_t equal to variance of x_t minus one plus variance of square root of one minus α_t into ϵ_t . This I can write as

variance of x_t minus one plus variance of a into ϵ_t , where a equal to square root of one minus α . Now, we know this will remain as same, x_t minus one plus a square variance of α . Okay, so this is becomes variance of x_t minus one plus α square means a square means one minus α , okay? And variance of t .

[15:06] Now, this variance of ϵ_t is one, so we can write it like this, okay? Now, so what happens is if I take the entire, uh, you you previous image, what happens there? The variance grows. So, this variance of x_t minus one, it keeps on growing. And after many steps, the variance would become very large, and the distribution will blow up. So, this is the problem if we take the complete image. So, that is complete image of the previous step.

[15:40] That is the reason we take a scaled version of the previous image. We have taken this expression x_1 equal to square root of α one x_0 plus uh square root of one minus α one into ϵ_1 . Now, if we expand it recursively, and we try to get what is my x_2 , so I will write it here. x_2 equal to square root of α two x_1 plus one minus α two ϵ_2 . Now, that I can write it as square root of α two for x_1 , what we'll write?

[16:25] α one square root x_0 plus square root of one minus α one ϵ_1 . This is my x_1 term. This is basically my x_1 term. plus one minus α two ϵ_2 . Okay? Then, I can write this one as square root of α one α two x_0 plus square root of α two one minus α one ϵ_1 plus square root of one minus α two ϵ_2 . Okay? Now, this is this term is actually having your signal information. So, this is uh This contains original signal. Correct? What is our original signal? That is my X_0 .

[17:32] And this term is a Gaussian noise term. This is also another Gaussian noise. Okay? So, can I write it like X_2 can be written as square root of α 1 α 2 X_0 plus sum of Gaussian noises. Okay? Now, if I make Let's consider α_2 bar equal to $\alpha_1 \alpha_2$. So, and let Z equal to addition of these two Gaussian terms, which is square root of α_2 into 1 minus α_1 into ϵ_1 plus square root of 1 minus α_2 into ϵ_2 .

[19:03] Okay? So, since ϵ_1 and ϵ_2 are independent Gaussians. Its covariance can be written as α_2 into one minus α_1 I It's covariance. Plus one minus α_2 I. That is if I do the calculation α_2 minus α_1 α_2 plus one minus α_2 I that is equal to one minus $\alpha_1 \alpha_2$. That is one minus α into I α_2 bar because this is my $\alpha_1 \alpha_2$ into I. Okay. So if we do it like this, we are getting X_2 equal to X_2 bar X_0 plus one minus α_2 bar into I.

[20:44] Okay. So this is the expression for X_2 . Okay. Now we will see the same thing I have generalized here. So X_T bar equal to product of S equal to one to T α S . Okay, so that α_2 bar I have generalized. And I have written it like this X_T bar.

Okay. Therefore, α_1 into α_2 into α_3 dot dot dot α_T can be written as square root of square root of α_1 into α_2 into α_3 dot dot dot dot α_T can be written as square root of this product term. Okay.

[21:33] Now, I can write the probability distribution for the forward diffusion process Q_{X_T} given X_0 equal to N_{X_T} . This is This is what has come there as the mean square root of X_T bar X_0 and this is the covariance. This is the mean and this is the variance of the covariance. Okay. Now, uh Till this one, we have seen how we have written the forward diffusion uh probability distribution equation for the forward diffusion process.

[22:23] Now, we will learn something which is called as reparameterization trick. What is that one? And it is a very very important concept in denoising diffusion probabilistic model. Okay. So, we will see. Now, suppose we have Z which is sampled from a Gaussian with mean μ and variance σ^2 . Then, instead of sampling Z directly, we can write Z equal to μ plus σ into ϵ . Okay, this is exactly what we were doing in the previous step also.

[23:06] So, the same thing here, we are, you know, reinforcing and we will see how with clarity what exactly it is coming out from this reparameterization trick and why it is important. Okay? Now, Z can be written as the mean plus σ , which is your standard deviation, into ϵ . ϵ belongs to $N(0,1)$, which is a distribution with a Gaussian distribution with mean zero and variance one, which is basically your standard Gaussian.

[23:43] If I apply this idea, then I can write q_{X_t} given x_0 equal to Uh if we apply this idea to this equation, which I had written before, I can write it. So, here my mean becomes This is my mean. And this is my variance term. Okay? Or covariance term. Okay? I can write it I can sample x_t as square root of that is the mean. This is basically your mean. Plus This is my σ . It's because square root has come. So, this is my σ .

[24:35] Into ϵ . So, what we are getting from this equation is I can get x_t given I know the initial image x_0 at any time instant t . So, this is a very important concept of denoising diffusion probabilistic model. So, what happens is this is basically your remaining signal strength and this is your accumulated noise. Okay? So, without the reparameterization if we require to find out X_T , I have to go from X_0 to X_T . In several steps, X_0, X_1, X_2, X_3, X_4 till X_T .

[25:26] It requires T diffusion steps. Okay? With this reparameterization technique, X_T can be obtained directly in one step for any random step T . Provided you know X_0 . X_0 is known to you because that we have given to the model. So, this is the way where in one step we can get get X_T . And we can avoid going through all the steps in the diffusion process. Okay? So, with this one I come to the you know, end of today's session.

[26:06] Uh let us before I conclude, let us have a quick look on what all we have studied today. So, we are studying the forward process, forward diffusion process in denoising diffusion probabilistic model. Which is a specific, you know, framework of diffusion probabilistic model. Okay? So, here also we have the forward process and the reverse process. The forward process is a known process. There is no learning that happens in the forward process. It is considered as a predefined Gaussian.

[26:40] And then what we have tried to see from there is uh how the mathematical expression of the diffusion of the mm, probability distribution is written in terms of uh mean and covariance. And the mean was a scaled version of your previous step image. Why that scaling is how that scaling helps, also we have seen if the scaling is not there, then the variance will blow up the variance will keep on increasing at each and every step and the distribution will blow up after several steps, okay? So, we have the mean as a a a a fraction of the previous image and we have a covariance term with that one.

[27:36] Now, we have seen how we can write that one in terms of uh you know, there are some mathematical steps by which we have seen the reparameterization trick, which is which allows us to get a sample X_T from X_T , T being any specific time step if from only the original image. It doesn't require for us to go through the entire T number of diffusion steps to get X_T , okay? So, there are couple of mathematical steps that we have done and we have seen how this reparameterization helps in you know, speeding up the process. So, at any instant of time you want to see how much of what is the noisy image that you are getting at T equal to 99, say.

[28:38] Then once you have X_0 with you, you can directly from X_0 you can find out what will be the form of X_{99} . Okay? So, that is the And it's a very important concept in DDPM or denoising diffusion probabilistic model. Okay? So, with this one, uh I come to the end of today's session. I hope you all have got some understanding about the forward diffusion process, the maths behind the probability distributions behind forward diffusion process. In my next class, um we will discuss about the reverse denoising diffusion reverse denoising process or reverse diffusion process in which the actual learning of the neural network happens. There will be parameters associated with that, you know, process.

[29:32] And we will see what are the mathematical formulations of the reverse process in my next class. Okay? Till that time, till we meet for the next session, keep learning, happy learning, and thank you all.